

Program : Diploma in Wood and Paper Technology	
Course Code : 6121B	Course Title: Paper board and specialty Paper
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To impart knowledge on speciality, coating, properties of Paper , Paper Board and Printability of paper

Course Prerequisites:

Topic/Description	Course code	Course Title	Semester
Knowledge of basic Mathematics		Mathematics I	1
Knowledge in Wood Processing,		Wood Processing	3
Paper properties		Paper Technology	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Contrast Paper and paper boards	13	Understanding
CO2	Explain coating of paper and specialty paper	15	Applying
CO3	Compare different properties of wood	15	Applying
CO4	Relate paperboard grades and printability of Paper	15	Applying
	Series Test	2	

CO – PO mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2							
CO3	3						
CO4							3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Contrast Paper and paper boards		
M1.01	Recall paper	4	Remembering
M1.02	Develop an idea about industrial paper board	4	Remembering
M1.03	Recall paper board	5	Understanding
Contents: PAPER BOARDS - Paper board - Introduction - Definition - Manufacturing process - Fiber analysis of paper. Multi-ply paper board - Definition - Manufacturing process - Cylinder type formers. Multiply board cross section with figure - Multiple Fourdriner board machine with figure - Multiply board properties. Industrial grade paper boards - Linear board - Corrugating medium - Construction paper. Liner board - Definition - manufacturing using Fourdriner machine - Pulping system for Liner board - Paper board treatments Grades and types of board - Electrical insulating boards - Stereo type mat - matrix board - Paper board - Corrugated containers - materials - Combined board structure - Corrugated adhesive - box board.			
CO2	Explain coating of paper and specialty paper		
M2.01	To explain on coating process	5	Understanding
M2.02	To illustrate coating equipment	5	Understanding
M2.03	To give an idea about specialty paper	5	Applying
	Series Test – I	1	

Contents:

COATING OF PAPER Coating- Coating equipment- Air Knife coater - Brush coater - Cast coating - Flexible blade coater - Gravure coater - Reverse roll coater - Coating Processes: Coating color preparation, coating techniques, multiple coating of paper, coating of board, drying of coated paper, process control and automation, rheology of pigment slurries Transfer roll coater.

SPECIALITY PAPER Grease proof paper - Cigarette paper - Tissue paper - Butter paper - Insulating paper - Filter paper - Newsprint - Sanitary towels.

CO3	Compare different properties of paper board		
M3.01	Explain properties of paper	4	Remembering
M3.01	Explain testing of paper	4	Understanding
M3.01	Explain paper boards	4	Understanding
M3.02	Develop an idea on optical properties	3	Applying

Contents:

PROPERTIES AND TESTING OF PAPER AND PAPER BOARDS Objectives of testing - Chemical testing of pulp - Physical testing of pulp - Testing procedure - Paper testing - physical, mechanical properties - Basis weight - Caliper - Density - Bulk - folding endurance - Tear - Burst - Tensile strength - Stretch - Softnes - Roughness.

OPTICAL PROPERTIES Brightness - Opacity - Gloss - water absorbency as per TAPPI

Characteristics of Specialty Grades: Electrical papers, absorbent papers, filter papers, special strong papers, release papers, copy and imaging papers, packaging papers, building papers, cigarette papers, and other functional papers.

CO4	To relate paperboard grades and printability of Paper		
M4.01	Explain printability of paper	5	Remembering
M4.02	Discuss Paper requirement for Letter press	5	Understanding
M4.03	Explain the idea on different grades of paper	5	Applying
	Series Test – II	1	

Contents:

Printability of paper- Printing- Printing process- Letter- Gravure- Lithography- Screen printing- Inks for letter press and lithography-

Paper requirement for Letter press and Lithography- Reprography- Xerox technology- Thermography- Gravure printing

Paperboard Grades: Carton boards classification and quality requirements for various applications, containerboards and linerboard

Text /Reference:

T/R	Book Title/Author
T1	K. W. Britt, Pulp and Paper Technology,CBS
R2	Robert. R.A &Higham ,Paper Board and Board- , CBS
R3	G. A. Smooke, Hand Book of Pulp and Paper Technology- Interscience
R4	Pulp and Paper Manufacture. Vol. I, II, III by Ronald G. Macdonald & John. N. Frankli

Online resources:

Sl No	Website Link
1	https://www.indiamart.com/proddetail/special-paper-board-22419138048.html
2	https://www.mondigroup.com/en/products-and-solutions/speciality-kraft-paper/
3	https://www.cappelen-group.com/en/Paper.htm
4	https://dir.indiamart.com/impcat/kraft-paper.html
5	https://dir.indiamart.com/impcat/brown-kraft-paper.html